

A Predictive Supervisory Layer for GPU Fleet Control

ARCHITECTURE, PUBLIC-REPLAY EVIDENCE, AND A HISTORICAL VALIDATION PROPOSAL

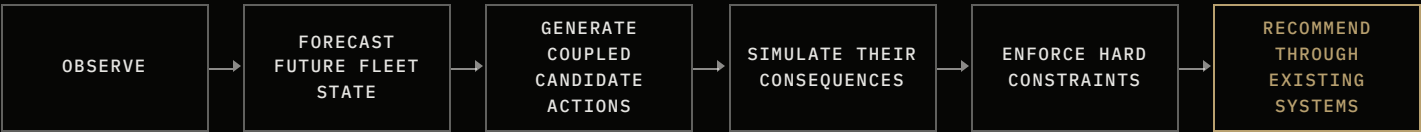
Aurelius does not replace Kubernetes, Slurm, serving schedulers, fleet-management systems, or existing control planes. It sits above them as a predictive supervisory layer.

Existing systems make sophisticated decisions at each layer. Aurelius tests whether those decisions can be improved by coordinating them against a shared prediction of future fleet state, simulating coupled alternatives before execution, and returning the best safe action through the existing control plane.

THE PROBLEM

At fleet scale, individually reasonable decisions about placement, capacity, routing, batching, prewarming, admission, migration, and serving policy interact. Their downstream effects arrive on different time horizons and cannot be evaluated reliably as isolated knobs. Some of the constraints that decide a decision's economics emerge only after it has executed.

THE ARCHITECTURE



Each control period, Aurelius advances a persistent forward model of the fleet, generates a bounded set of high-value coupled candidate actions, evaluates each candidate on a read-only simulated future, rejects anything that fails a hard SLA, capacity, power, policy, or safety gate, and returns the best feasible action through the systems that remain authoritative.



THE EVIDENCE AND THE NEXT STEP

Public replay produced +723.7 percent, reported as +724 percent, average SLA-safe goodput per dollar under the specifically disclosed uncapped high-load benchmark, with 84.5 percent fewer GPU-hours and better SLA outcomes in all three market windows. Page 5 reports the full table, the baseline definition, and the scope.

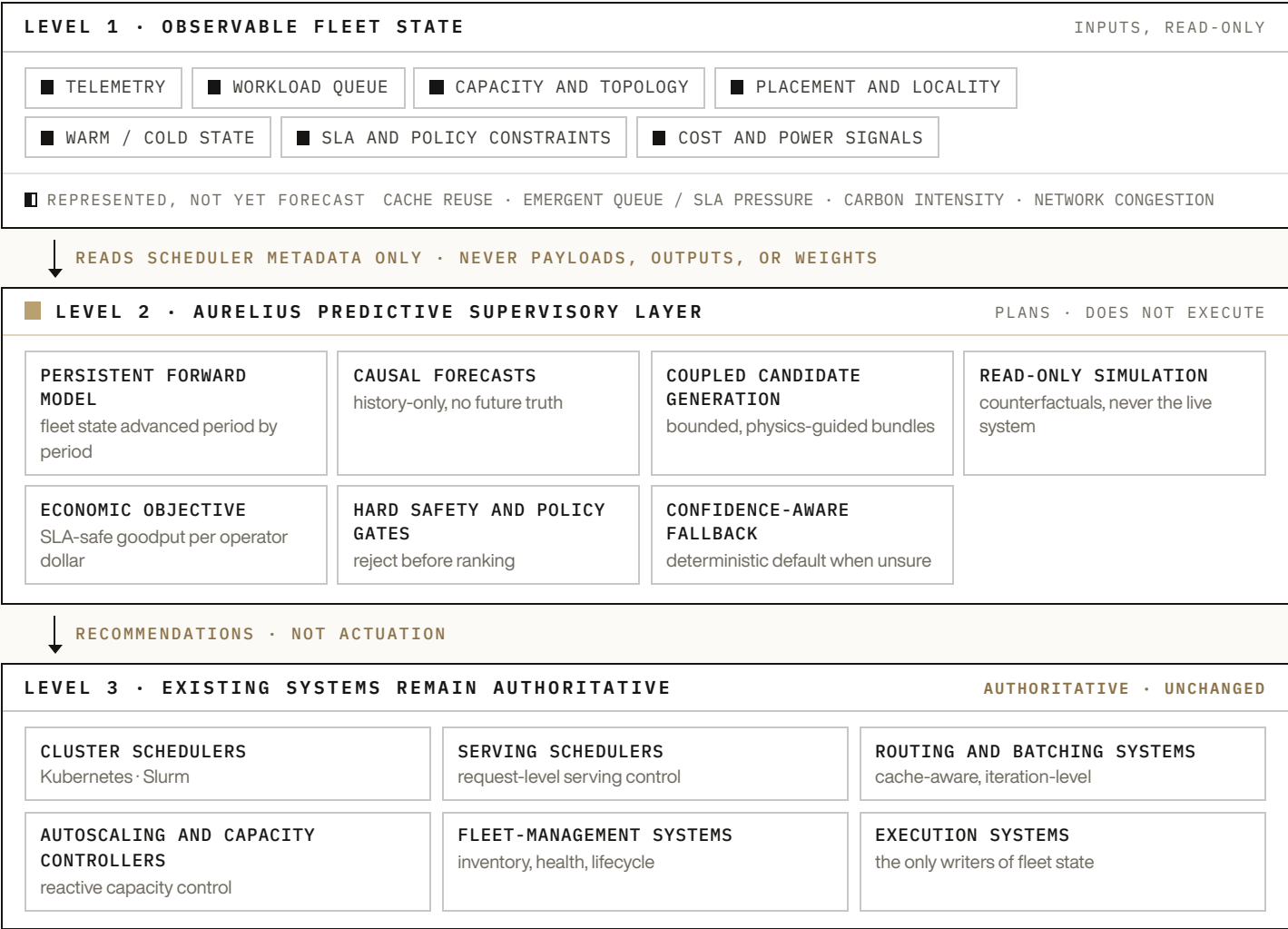
+724% AVG SLA-SAFE GOODPUT PER DOLLAR · UNCAPPED PUBLIC REPLAY	84.5% FEWER GPU-HOURS AT IDENTICAL REPLAYED LOAD	~1.5M REPLAYED REQUESTS · THREE SELECTED MARKET WINDOWS	0 MEASURED SEARCH REGRET WHERE ENUMERATION WAS TRACTABLE
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SCOPE This is a **simulated, load-dependent result from selected public-trace windows**. It does not establish expected savings for a particular fleet. The proposed next step is a **bounded, read-only historical replay using anonymized scheduler metadata** (page 6).

THE BOUNDARY

A supervisory layer, not a replacement control plane

The architecture is easiest to state as a boundary. Aurelius consumes the state that fleet systems already expose, plans against a prediction of where that state is going, and returns recommendations to the systems that already own execution.



THE COUPLING PROBLEM

A fleet does not face one scheduling problem

A large fleet faces thousands of interacting decisions made at different layers and over different horizons. A locally correct action can change the value or feasibility of a later action elsewhere in the system. The effects also land on different clocks: batching within the current window, capacity and prewarming over subsequent periods, placement and migration well beyond them.

DECISION SURFACE	FIRST-ORDER EFFECT · WITHIN THE PERIOD	COUPLED EFFECT · LATER, ELSEWHERE	COUPLING
BATCHING	raises throughput for the current window	changes queue delay and tail-latency risk	
PREWARMING	reduces cold-start risk for forecast arrivals	consumes capacity that other work needs	
PLACEMENT	improves locality for the placed workload	reduces flexibility for later workloads	
CAPACITY	valuable when added ahead of a predicted surge	wasteful when held continuously	
MIGRATION	escapes predicted congestion	consumes bandwidth and capacity while in flight	
ADMISSION	relieves immediate queue pressure	alters SLA risk in later periods	
ROUTING	improves cache reuse and locality	concentrates load on specific replicas	
PRECISION · CLOCK	changes per-request execution time	changes GPU consumption and power headroom	

FIG.02 · GOLD LINKS MARK DEFERRED EFFECTS THAT LAND ON ANOTHER SURFACE'S DECISION PROBLEM

THE VALUE HYPOTHESIS

The hypothesis is not that existing controllers are unsophisticated. It is that the best action at one layer may depend on future conditions and decisions at several other layers.

WHERE CURRENT-STATE OPTIMIZATION STARTS TOO LATE

Some important costs and constraints emerge only after a placement or capacity decision has already been made: capacity and power headroom, congestion and topology pressure, demand, and the wholesale price of electricity, which on public day-ahead markets can swing several-fold within a day. Once the decision lands, those costs are largely locked in. A forward model of future fleet state allows those conditions to influence the decision before execution rather than explain it afterward. No single lever is appropriate for every workload: candidates are generated per operating regime and workload class, and hard policy gates reject any bundle that is out of bounds for the work it would touch, before ranking.

THE CONTROL LOOP

Forecast, simulate, constrain, decide

One loop, six stages, each control period. The loop is deliberately conservative: prediction and search are allowed to be ambitious because nothing they produce reaches execution without passing the gates.

01 PERSISTENT FORWARD MODEL

A stateful model advances the fleet through time rather than treating the system as a snapshot of free GPUs. It carries server and rack capacity, placement and locality, queue dynamics, replica warm and cold state, migrations in flight, and the operator cost basis, including warm-hold and migration penalties. Inputs are tagged by provenance, so a modeled value is never treated as a measured one.

02 CAUSAL FORECASTS

Forecasts of arrival rate, request characteristics, queue dynamics, and infrastructure and pricing conditions, including wholesale electricity price, use only information available at the current decision point. No future arrivals and no future prices leak into any decision, in planning or in evaluation.

03 PHYSICS-GUIDED CANDIDATE GENERATION

A bounded set of plausible, high-value coupled bundles is generated from the operating regime: timing, placement, routing, capacity, batching, prewarming, admission, migration, precision, and clock policy are proposed together rather than tuned as independent knobs. Known-strong candidates are always retained, so a valuable option is not silently excluded.

04 READ-ONLY COUNTERFACTUAL SIMULATION

Each candidate is evaluated on an isolated simulated future advanced by the forward model. Candidate evaluation never mutates the live system, and scoring is repeatable and auditable after the fact.

05 OBJECTIVE AND HARD GATES

The objective is SLA-safe goodput per operator dollar: SLA-meeting requests over infrastructure cost, with energy at the prevailing market price plus warm-hold and migration cost, and a forecast-uncertainty penalty in the same economic terms. Hard capacity, SLA, power, policy, residency, and safety constraints reject infeasible candidates before ranking; a Pareto gate accepts a recommendation only if it beats the baseline without regressing SLA.

06 RECOMMENDATION AND FALLBACK

The best feasible candidate is returned as a recommendation through the existing control plane. When forecast confidence is insufficient, Aurelius falls back deterministically to the configured safe default rather than acting on a low-confidence forecast.

The optimizer cannot select a decision it never evaluates.

Which candidates enter the search is a first-class engineering problem, measured under a regret audit, not an afterthought.

WHY CANDIDATE GENERATION IS A FIRST-CLASS PROBLEM

The sharpest finding in the current benchmark cycle was not about forecasting. With the forward model, simulator, objective, and safety gates all unchanged, enabling more decision surfaces initially made results worse. The cause was containment: the candidate set had collapsed around a single surface, so the bundle that actually wins an expensive electricity window (lower precision, aggressive batching, and a higher clock) was never in the set being evaluated. Restoring the physics-guided candidate set recovered most of the opportunity; letting the coupled search reach combinations a fixed grid does not contain recovered the rest. Where exhaustive enumeration was tractable, the bounded search matched it exactly, at roughly 40 evaluations per decision. The residual gap to a forecast oracle, about 4.5 percent, is forecast quality, not search.

CANDIDATE SET	Δ GOODPUT/\$ VS BASELINE	SLA GATE	SEARCH REGRET VS EXHAUSTIVE
Single surface (prior)	-4.47%	fails	173.5% forfeited
Physics-guided candidate set	+100.48%	holds	30.3%
Physics-guided set + coupled search	+161.31%	holds	0%

Benchmark V1, primary market window. Same simulator, objective, cost model, and Pareto gate in every row, so the difference is attributable to search and candidate generation. This is a mechanism finding under the frozen continuity harness, not the headline benchmark.

THE RESULT

A large signal, with its scope attached

Replaying the full selected public-trace windows with the per-period request cap removed produces the largest verified result in the program. It is reported the way it should be read: baseline defined, load disclosed, scope attached.

The comparison policy is a constructed production-class composite: one deterministic, causal policy assembled from established production serving techniques. Continuous, iteration-level batching^[1,2]; SLA-aware ordering with deadline protection and class-based admission under pressure^[3]; KV and prefix-cache-aware routing^[4]; topology-aware placement^[5]; and backlog-driven autoscaling with warm-pool headroom^[6,7]. It sees no future arrivals and no future prices. It is a public-replay composite, not any specific operator's internal scheduler.

MARKET WINDOW	AURELIUS GP/\$	COMPOSITE GP/\$	Δ GP/\$	SLA VIOL. AURELIUS	SLA VIOL. COMPOSITE	GPU-H AURELIUS	GPU-H COMPOSITE
PJM · expensive window	1,041,842	130,539	+698.1%	0.0023	0.0382	62.2	399.9
ERCOT · expensive window	1,064,602	130,178	+717.8%	0.0013	0.0447	51.0	323.7
CAISO · expensive window	1,111,915	130,000	+755.3%	0.0018	0.0458	60.0	391.5
Mean of per-market deltas / totals			+723.7%			173.1	1,115.1

gp/\$ is SLA-safe goodput per operator dollar: SLA-meeting requests divided by infrastructure cost (energy at the prevailing day-ahead market price, warm-hold, and migration cost), read directly from run output. Replayed requests: PJM 576,912; ERCOT 442,716; CAISO 529,621; total 1,549,249. The +723.7% headline, publicly rounded to +724%, is the unweighted mean of the three per-market deltas; total GPU-hours are 84.5% lower at the same served load. Workload: public LLM inference traces^[8,9]; prices: public day-ahead hourly wholesale series^[10].

■ SCOPE AND INTERPRETATION

READ BEFORE QUOTING THE NUMBER

- Approximately 1.5 million replayed requests across three selected high-load windows; not the full lifetime of the traces.
- Identical selected windows and identical load for Aurelius and the constructed composite.
- No policy sees future arrivals or future electricity prices. Measured per-request execution time is used rather than predicted, so there is no oracle.
- Simulated replay, not a production deployment.

- The baseline is a constructed public-replay composite built from established production serving techniques. It is not a specific operator's internal scheduler.
- The percentage is load-dependent. Part of the gap is a fixed-policy composite degrading under heavy uncapped load while Aurelius adapts its economic posture to it.
- Modeled precision and batching behavior affect the magnitude.
- The correct interpretation is evidence that the architecture warrants validation, not a prediction of savings for a particular fleet.

FROZEN CAP-CONTROLLED CONTINUITY BENCHMARK	V1 WINDOW	Δ GOODPUT/\$	SEARCH REGRET
Benchmark V1 is the prior frozen harness: a small fixed per-period request cap, scored against the strongest SLA-aware baseline. Under that harness, Aurelius delivered +147.85% to +191.41% versus the strongest SLA-aware baseline across the three reported market windows, with zero measured search regret where exhaustive evaluation was tractable.	PJM	+161.31%	0
	ERCOT	+191.41%	0
	CAISO	+147.85%	0
Different baseline and different load from the uncapped benchmark above. The two results are reported separately and never mixed or averaged.			

THE PROPOSAL

A bounded, read-only historical evaluation

The next question is not whether the public benchmark reproduces exactly on another fleet. It is whether predictive cross-layer planning identifies meaningful, causally defensible counterfactual improvements on real historical fleet behavior.

One representative workload, cluster, or bounded historical window	One technically appropriate operator or internal owner	One predetermined evaluation protocol, fixed before data moves
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POTENTIAL INPUTS · ANONYMIZED SCHEDULER METADATA

– Job or request arrival time	– The historical scheduler action
– Workload class	– Configured hard constraints
– Requested and allocated GPU resources	– Warm and cold replica state, where available
– Queue, start, completion, and deadline information	– SLA outcome
– GPU type and capacity state	– Optional internal cost or power proxy
– Placement and topology information, where relevant	

The precise schema would be minimized to the selected evaluation. Nothing outside it is requested.

NOT REQUIRED

☐ Customer identity
☐ Request payloads
☐ Model outputs
☐ Proprietary weights
☐ Unrestricted infrastructure access
☐ Production actuation
☐ Immediate integration into live systems

Historical replay precedes any shadow-mode recommendation, and shadow mode precedes any optional production integration. Every stage remains under operator control.

EVALUATION OUTPUTS

– Faithful replay of the historical baseline	– Simulator-fidelity checks against held-out telemetry
– Aurelius counterfactual recommendations on the same windows	– Sensitivity analysis
– Matched SLA, GPU-hour, utilization, and economic comparisons, where valid	– Failure cases and negative findings, reported as such
– Attribution by decision surface	– A clear recommendation on whether shadow-mode evaluation is warranted

THE ASK

Identify one technically appropriate owner and one bounded historical dataset on which the predictive-planning hypothesis can be evaluated.

Public replay establishes a testable signal. Historical operator telemetry is the next step required to determine whether the architecture generalizes.

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FOUNDER · AURELIUS

The complete benchmark methodology, citations, limitations, and version history are available in the supporting full technical report.

SELECTED SOURCES AND METHODOLOGY

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